

TOSHAL ZAMBARE

AI / ML Engineer | Python | Deep Learning

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SUMMARY

Final-year AI & Data Science student (CGPA 8.7/10) who builds real AI systems. Fast learner, hackathon winner, and Cloud Lead at Google Developer Groups on Campus.

TECHNICAL SKILLS

Languages: Python, C, C++, Dart

AI / ML: LLMs, LangChain, RAG, NEAT, TensorFlow, Keras, HuggingFace, Sentence-Transformers, OpenCV, CNN, Prompt Engineering

Data Science: NumPy, Pandas, Scikit-learn

Backend & Databases: FastAPI, Flask, Celery, Redis, PostgreSQL, Qdrant, MinIO

Tools: Docker, AWS, Google Cloud, Git, OOP, Data Structures and Algorithms

INTERNSHIP

Deep Learning and Full Stack Intern | AI Leela, Nashik

Dec 2025 - Feb 2026

Knowledge Hub & Data Ingestion System

- Architected an event-driven, containerized backend (**FastAPI**, **PostgreSQL**, **Redis**) integrating OAuth APIs to perform automated delta-syncs across **Google Drive**, **OneDrive**, and **Gmail**.
- Engineered an asynchronous, multimodal **data ingestion pipeline** using horizontally scaled **Celery workers** to extract and process PDF, XLSX, and MP4 files via **LangChain** and **Whisper**, archiving raw assets in **MinIO**.
- Designed a highly accurate **Two-Stage RAG** search engine - generated 384D vector embeddings via **SentenceTransformers**, indexed in **Qdrant**, and applied a **Cross-Encoder** for strict semantic reranking.
- Implemented an **LLM Factory Pattern** supporting **OpenAI**, **Anthropic**, and **Ollama**, using **Server-Sent Events (SSE)** to stream low-latency chat responses and real-time analytics to a **React/Redux** frontend.
- This project is now live and actively used locally by AI Leela in production.

PROJECTS

AI NeuroEvolution Agent - NEAT Flappy Bird

2025

Python, Pygame, NEAT, Neural Networks, Matplotlib

- Engineered a reinforcement-learning environment where neural networks learn to autonomously play a game using the **NEAT (NeuroEvolution of Augmenting Topologies)** algorithm.
- Developed a real-time, **dual-mode visualization dashboard** in Pygame to monitor live network topology evolution and generation-over-generation mutations.
- Implemented **dynamic difficulty scaling** and pixel-perfect collision detection, forcing agents to adapt to moving obstacles and variable gaps.
- Built a custom **mutation-tracking engine** comparing genome snapshots, translating weight and node adjustments into human-readable logs.

Voice-Interactive AI NPC System

2025

Python, Flask, Unity, C#, Google Gemini API, Vosk, pyttsx3

- Architected an end-to-end **voice-driven AI communication system** integrating a **Unity 3D** client with a custom **Python microservice (Flask REST API)** for real-time, bidirectional, zero-blocking NPC conversations.
- Implemented offline **Speech-to-Text (Vosk)** and **Text-to-Speech (pyttsx3)** pipelines with a **thread-safe multiprocessing architecture**, achieving sub-second latency without paid cloud APIs.
- Integrated **Google Gemini API** in the C# client to dynamically generate context-aware NPC dialogue based on transcribed player speech.

LEADERSHIP AND HACKATHONS

1st Place - Cricketthon Hackathon, GDG Nashik (2026) | **3rd Place** - HackFusion Hackathon (2025) | **Cloud Lead**, GDG on Campus at MET | **Champion Tier**, Google Cloud Arcade | **Contributor**, Hacktoberfest

EDUCATION

B.E. in Artificial Intelligence and Data Science (Pursuing) | CGPA: 8.7/10

Expected 2027

Higher Secondary (Class 12) | Ord. Factory High School and Jr. College | 72.67%

2023